

# ● Epoch vs Iteration in Deep Learning

*The most confusing terms explained simply*

Understand how models learn – one batch at a time.

#DeepLearning #MachineLearning AI #NeuralNetworks

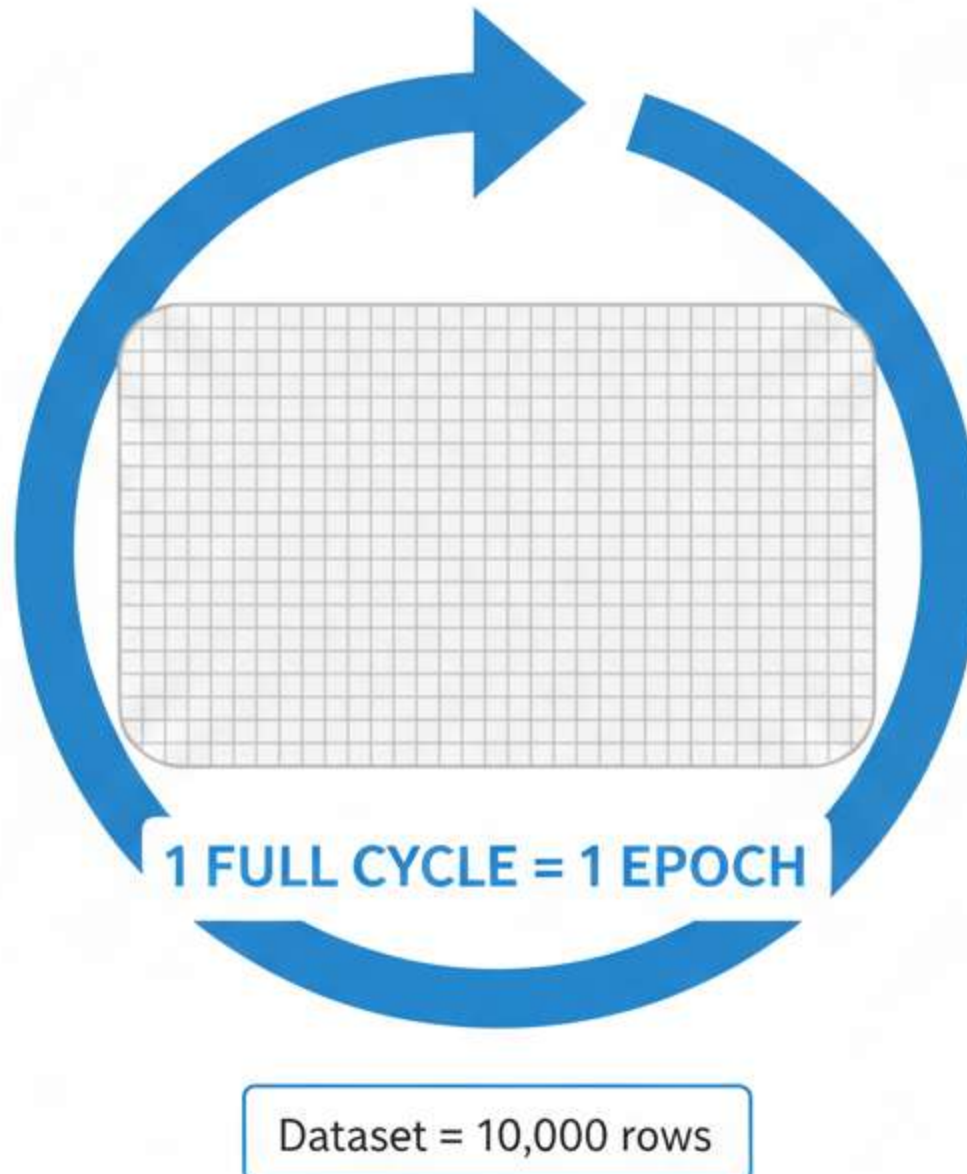
# ● Epoch = One Full Pass Through Data

- Model sees the entire dataset once
- Contains many iterations inside it
- More epochs = more learning cycles

## Example:

Dataset = 10,000 rows

Epochs = 5 → model sees each row 5 times



# ● Iteration = One Batch → One Weight Update

- Happens inside an epoch
- One batch forward pass + backward pass

## Example:

Dataset = 10,000 rows

Batch size = 100

→ 100 iterations per epoch



Each batch = 1 iteration

# Simple Analogy

## ● Epoch = Full Workout

(Your entire exercise plan done once)



## ● Iteration = One Set

(One set inside the workout)





## Quick Formula + Final Summary

**Iterations per Epoch** = Dataset Size  
/ Batch Size



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### Summary Points:

- Epoch = full dataset cycle
- More iterations → more weight updates
- More epochs = better learning (until overfitting)

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